ETH zürich

Mobile Computing

Specialized Applications
Consumer and Medical Devices

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems - Specialized Applications
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Overview

- Motivation - why Mobile Computing?
  - Digital therapy management
- The human body and the pancreas
  - Turning food into energy
  - The role of insulin and glucagon hormones
- Insulin Regulation Disorder (IRD)
  - Type 1 and Type 2 Diabetes
- Artificial Pancreas
  - Automated Insulin Delivery
  - Closed Loop Control

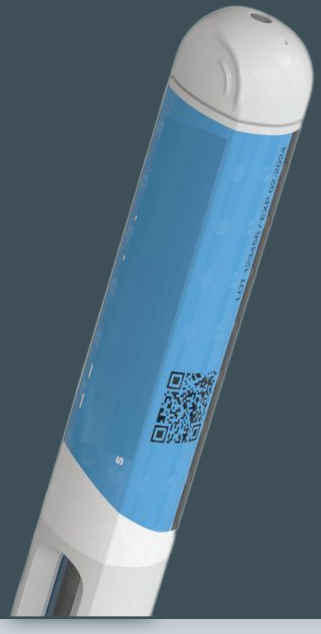


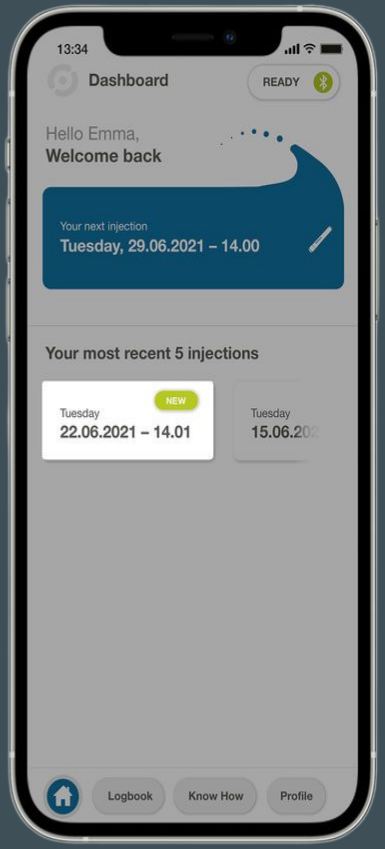
Mobile Computing for Medical Technology

- Automated medicine delivery is accepted way to administer medication
- Smartphones can augment control systems and offer additional (unlimited) computation
  - Device, edge, and cloud computing
- Therapy management increasingly important
  - Health care professionals in the loop
  - Adherence to a therapy plan, and efficiency can be traced

Digital Therapy Management

- Wireless data
- VLC for activation





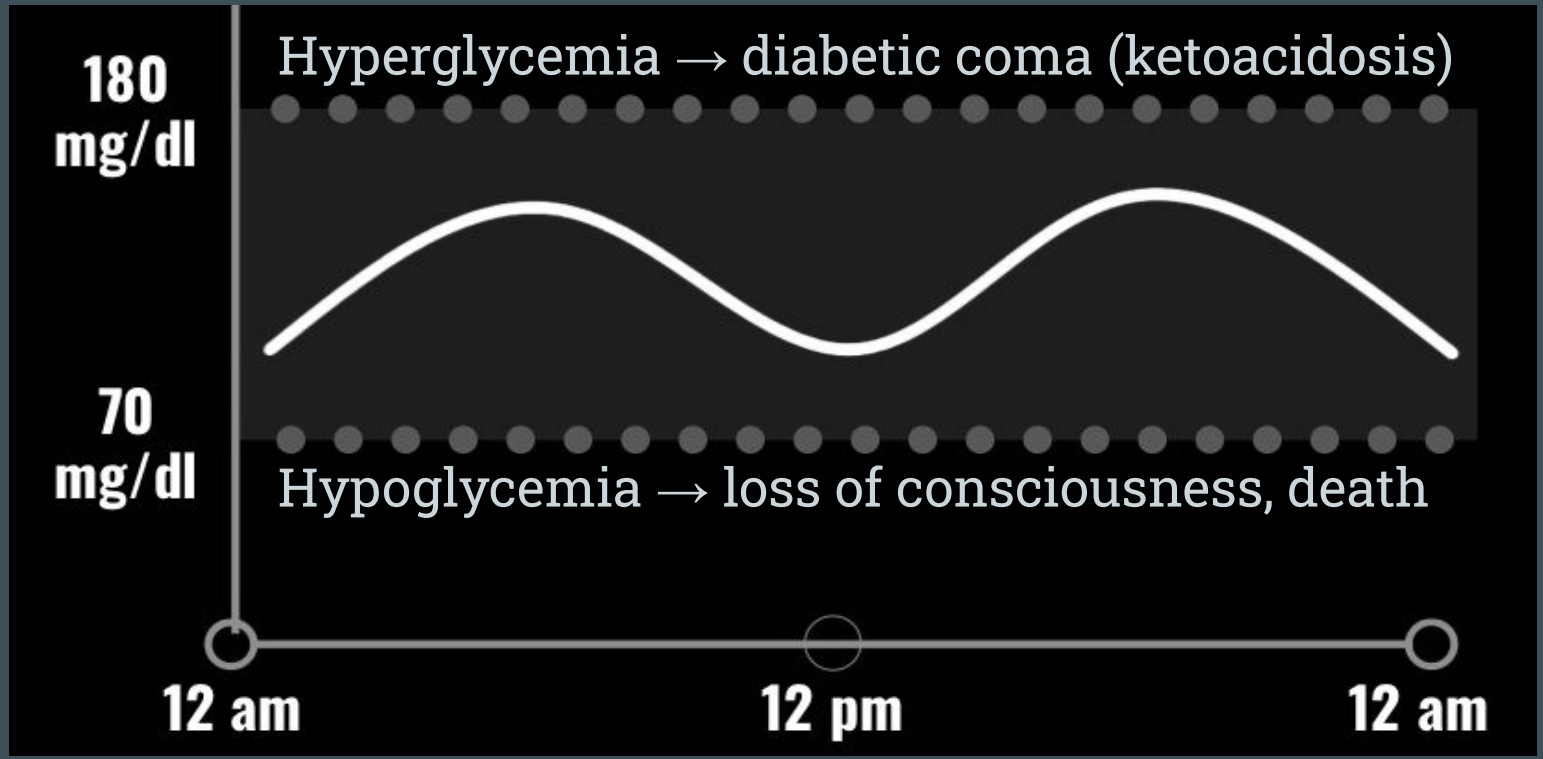
Digital Therapy Management

- Adherence management of self-medication
- Automated injection logging
- Assist patients, caregivers and healthcare professionals



The Human Body and the Pancreas

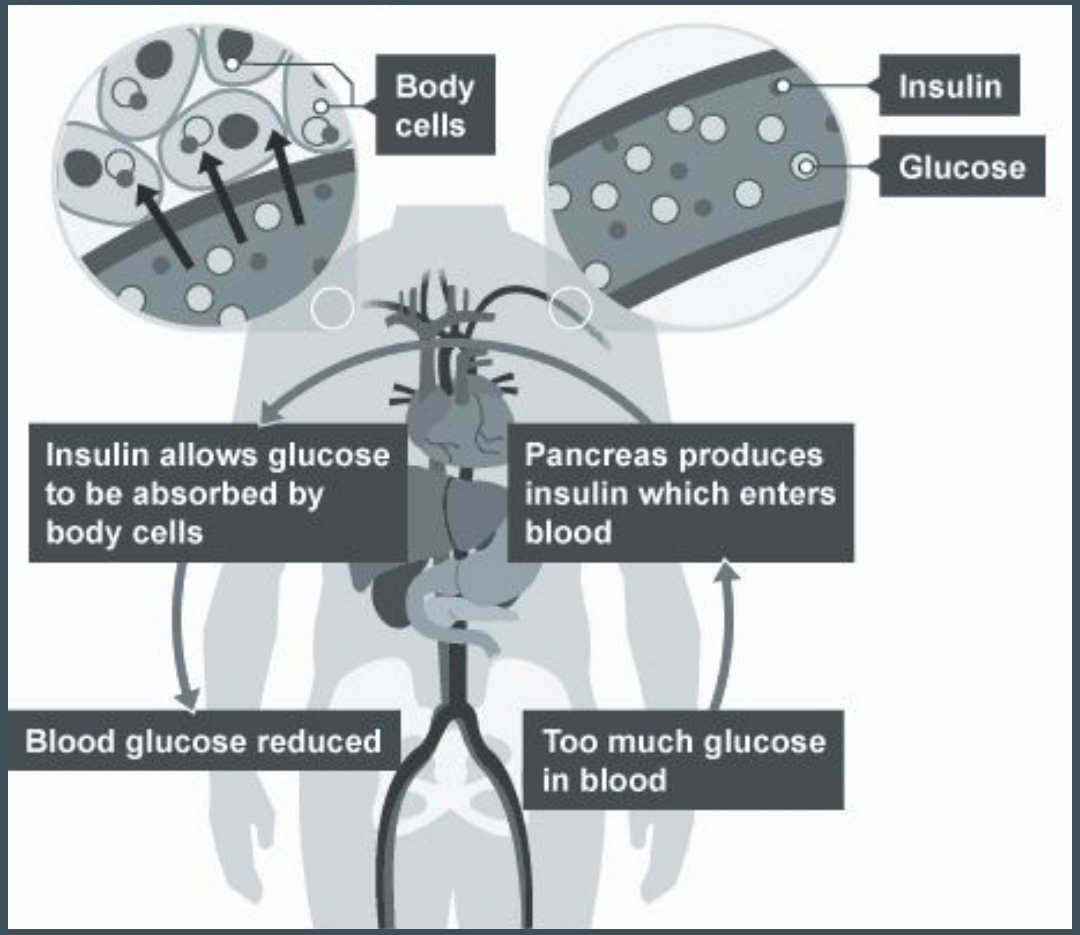
- A healthy pancreas regulates the time in range



The Human Body and the Pancreas

- The pancreas (“Bauchspeicheldrüse”) is an organ
- The pancreas generates two hormones: Insulin and Glucagon
  - A hormone is a signaling molecule in a multicellular organism, transported to distant organs to regulate physiology and behavior
  - Insulin helps cells in the body to take up and consume glucose (=sugar) from blood, which lowers blood glucose levels
  - Glucagon causes the liver to release stored glucose, which raises blood glucose levels (= converting glycogen into glucose)

The Human Endocrine System



bbc.co.uk/bitesize/guides/z8t47p3/revision/1

Insulin Regulation Disorder (IRD), Diabetes

- A group of metabolic disorders characterized by high blood glucose (hyperglycemia)
  - due to defects in insulin secretion, insulin action, or both
- Considered a chronic, long-lasting health condition
  - affects how a body turns food into energy
  - Type 1: pancreas produces very little or no insulin
  - Type 2: not enough insulin, or body resistance

Insulin Regulation Disorder (IRD), Diabetes

- Patients with type 1 diabetes and some patients with type 2 diabetes inject insulin to regulate their blood glucose (sugar)
  - Some might inject glucagon as well
- Blood sugar must remain in range
  - to lower risk of long-term complications: blindness, kidney failure, cardiovascular disease

Living with IRD

- Users have to periodically and manually test their blood glucose levels with a glucose meter, calculate insulin doses
- They administer the necessary insulin doses to lower the blood glucose (3…4 times per day)
  - Bolus: fast-acting insulin, typically taken with a meal
  - Basal: slow-acting insulin to cover the glucose output of the liver, typically taken once a day
- Glucagon injection treats low blood glucose
  - After loss of consciousness (otherwise, just eating sugar)

Continuous Glucose Monitoring, CGM Sensor

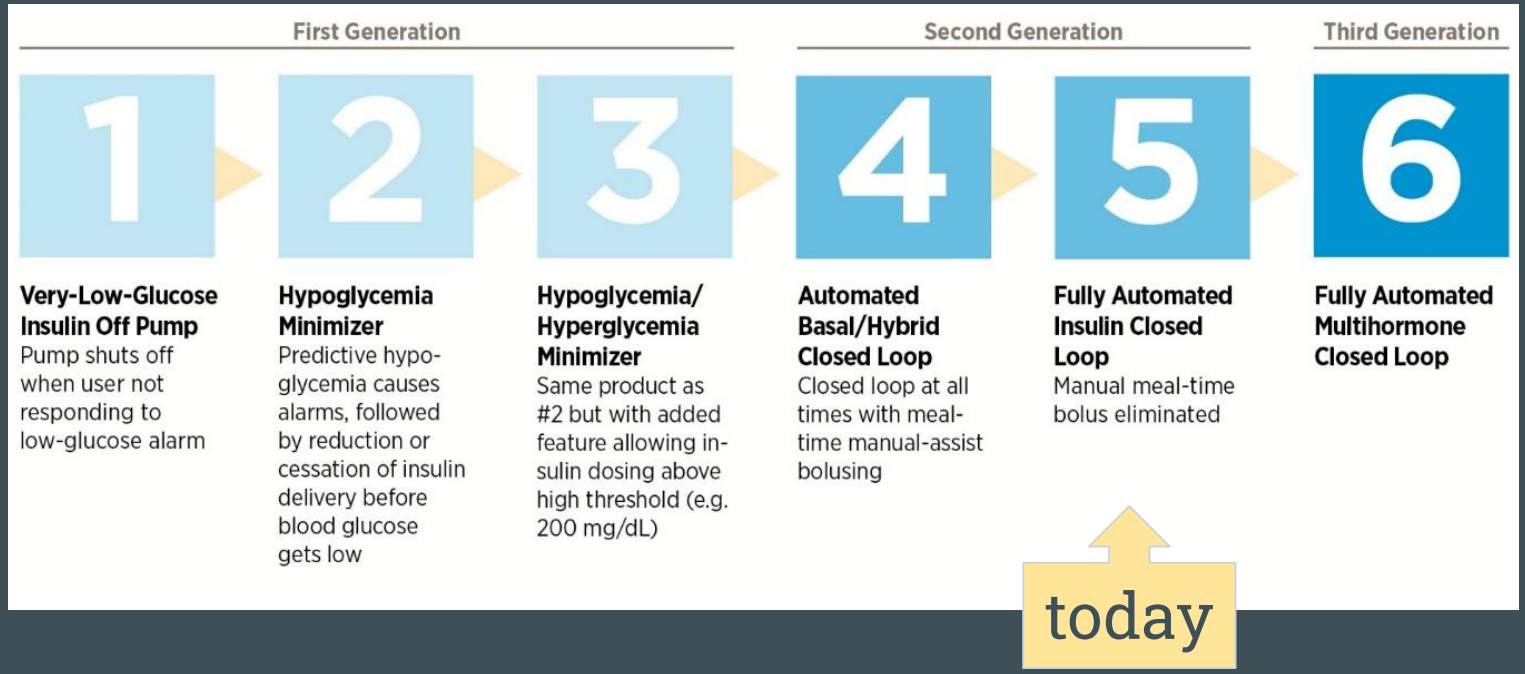


Artificial Pancreas

- Continuous glucose monitoring system (CGM) & Smartphone
  - Sometimes: Blood glucose meter to calibrate the CGM
  - Insulin infusion pump for subcutaneous (below the skin) injection
- Open-loop: most commercial systems
  - Users read the calculated recommendations on the smartphone
  - Users confirm or change the amount of insulin
  - Users inject their dosis
- Closed-loop: no user interaction, automated delivery

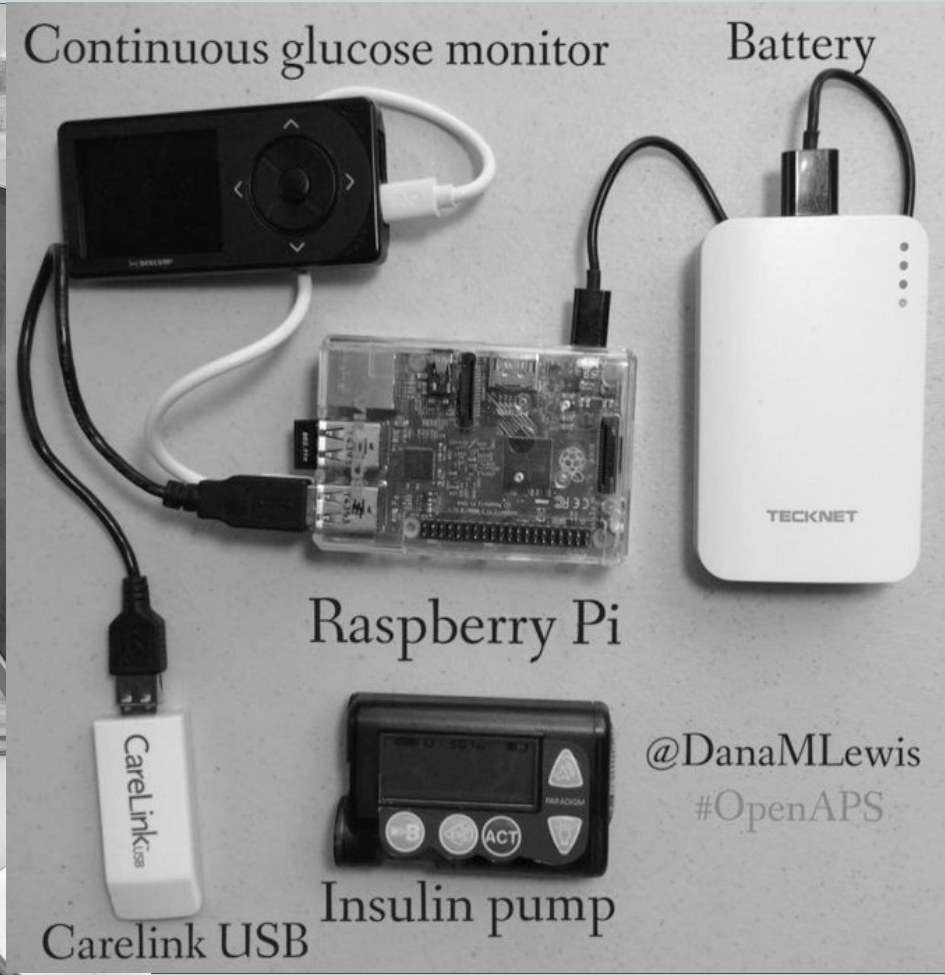
Generations of Artificial Pancreas

- Growing capability (and complexity) of available systems



OpenAPS: “Open Artificial Pancreas System”





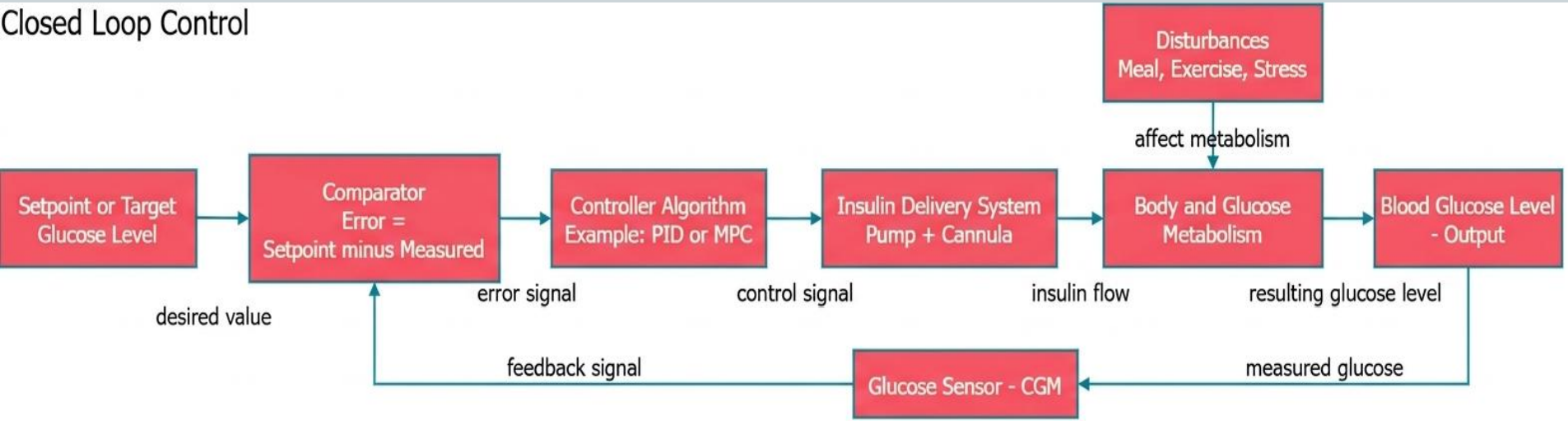
Do-it-Yourself System, OpenAPS

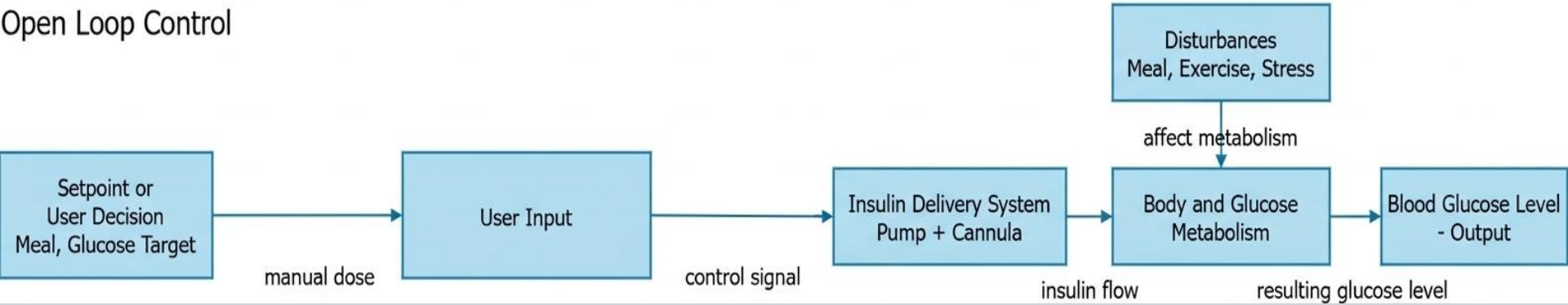


Control Algorithm (openaps.org/reference-design)

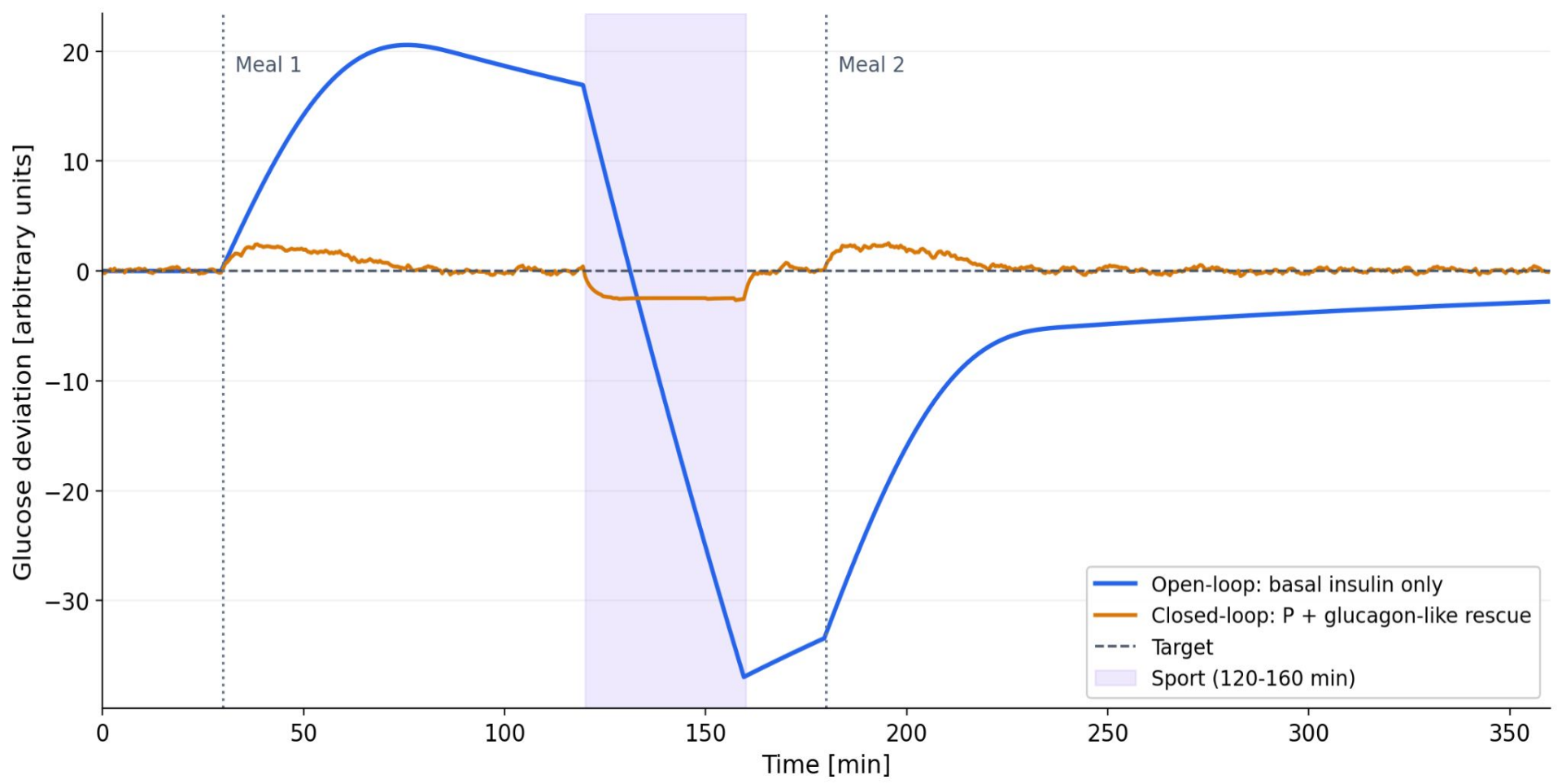
- The OpenAPS algorithm automates insulin delivery with insulin pumps and continuous glucose monitors (CGMs)
- At its core, it relies on a known control approach to manage and predict blood glucose (BG) levels
  - Adjusts insulin dosing based on glucose readings & predictions
  - Keeps blood glucose within target range, if possible
- Designed to be safe and conservative
  - Features like "safety caps" to limit how much delivered at once

Control Loop





Open- vs. Closed-Loop Artificial Pancreas



Challenges

- Bluetooth 5 security might not be sufficient
  - Might use Ultra Wideband instead
- Reliability of smartphones is difficult to guarantee
  - Battery, background apps, roaming, wireless connectivity
- Augmenting today's deterministic (PID) control algorithms with ML / AI algorithms is not supported by regulators
- 5G connectivity, together with edge computing (computing in the network) might be helpful

Exercise: Control and Coordination

- Open in your course folder: 02_Notebooks/04_Mobile_Algorithms/04.1_Control_and_Coordination_in_Mobile_Systems.ipynb
  - Run the cells and answer the questions.

